

LESSON 10.6

SKETCHING GRAPHS OF FUNCTIONS



LESSON INTRODUCTION

MA.8.F.1.3 Analyze a real-world written description or graphical representation of a functional relationship between two quantities and identify where the function is increasing, decreasing, or constant.

LEARNING TARGETS

- I can determine the independent and dependent variables being related in a functional relationship.
- I can sketch a graph from a description of a functional relationship.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

N/A

ESSENTIAL QUESTIONS

- How do you determine the independent and dependent variables in a functional relationship?
- How do you sketch a graph from a description of a functional relationship?
- Does the dependent variable increase, decrease, or remain constant as the independent variable increases?

LESSON NARRATIVE

10.6.2 Activity is a student-centered exploration that is the primary driver of instruction in this lesson. Students engage with videos or screenshots of four different stories. The 10.6.2 Activity positions students to make connections between what they are visually seeing happen and how they might represent that story using a graph and coordinate plane. Students can practice engaging with vocabulary that incorporates dependent, independent, axes, rates, and other mathematical language while still informally making sense of math in real-world contexts by sketching and interpreting their graph.

COMPONENT SUMMARY

10.6.1 WARM-UP (~5 MIN.)

Identifying independent and dependent variables

10.6.2 ACTIVITY (25-40 MIN.)

Watching videos and viewing screenshots of four stories before sketching graphs

10.6.3 TRY IT (10-20 MIN.)

Interpreting the graph of a bike race between two competitors

10.6.4 WRAP-UP (~5 MIN.)

Open-ended application for interpreting unique sketches

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms: N/A

MATERIALS

- (Optional) Note cards
- (Optional) Chart paper
- (Optional) Markers

ADDITIONAL PREPARATION

- Activity 10.6.2 contains a Desmos activity that accompanies the question prompts

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may confuse or invert the independent and dependent variables (or axes).
- Students may interpret lines or curves on a graph as directions (i.e., a straight line as traveling straight).
- Students may have a difficult time representing the piecewise and nonlinear functional relationships.

THINGS TO CONSIDER

- Prepare Desmos Activity Builder as the primary facilitator of instruction for 10.6.2 Activity and this lesson.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

At the end of the lesson, ask students to look back over the graphs from the lesson. Ask them what they notice about the independent variables of the graphs throughout the lesson (time was the independent variable for all of the functions). Then ask students what they wonder about independent and dependent variables. This language routine will help students synthesize and articulate their understanding.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 Real-World Contexts

Throughout the lesson, students connect mathematical concepts to everyday experience. They model real-world scenarios with graphs and also analyze and describe graphs that represent real-world functional relationships. Students must critically think about the how the independent variable impacts the dependent variable.

DIFFERENTIATE INSTRUCTION

Notice the real-world contexts of each story. Consider offering extra credit for students who create their own Graphing Stories at home with their own lived experiences.

10.6.2 Activity may be more difficult if students are interpreting the screenshots without the video. If your students do not have access to technology, consider playing each video for the whole group instead. Slides 10-12 in the Desmos Activity Builder provide the correct answers, so if students are using the videos independently, restrict the slides so they do not see the answers too soon.

Consider having students put on their best commentary voice for 10.6.3 Try It and submit a FlipGrid or verbal description of the race as if they were an ESPN anchor.

10.6.1 WARM-UP: DOG RUN

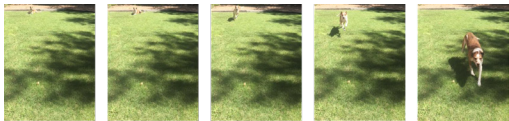
DIFFERENTIATE INSTRUCTION

Monitor for common misconceptions or trends in responses. Consider using these informal observations to inform partners or small groups for 10.6.2 Activity.

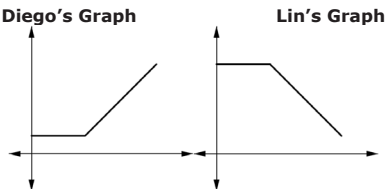


10.6.1 Warm-Up: Dog Run

Below are five pictures of a dog that were taken at equal intervals of time.



Diego and Lin drew different graphs to represent this situation.



1. They both used time as the independent variable. What represents the dependent variable on each person's graph?

Diego's Dependent Variable	Lin's Dependent Variable
<i>Distance from the back of yard</i>	<i>Distance from the camera</i>

2. Explain your reasoning.
Answers will vary. In Diego's graph the distance is increasing, meaning the dog is traveling farther away from the point of reference, the back of the yard. In Lin's graph the distance is decreasing, meaning the dog is coming closer to the point of reference, the camera.



10.6.2 Activity: Graphing Stories



In this activity, several videos will be shown. If you are not able to view the videos, screenshots and brief descriptions are included for your reference.

Story 1 – Playground Slide	
Screenshot	Description
	A man stands at the bottom of the ladder before climbing. He takes about 5 seconds to climb the ladder.
	The man walks across the platform at the top of the ladder to the top of the slide.
	Eight seconds after the man started to climb the ladder, he takes a seat at the top of the slide.
	He begins to slide down the slide at the 12-second mark.
	After 15 seconds have passed, the man is standing at the bottom of the slide.

- What are some different measurements that are changing over time in this story?
Some different measurements changing over time in this story are the man's vertical distance from the ground and the man's horizontal distance from the bottom of the steps.

- Sketch the graph of the relationship between the height of the man's waist above the ground, in feet, and time, in seconds.
Sketches will vary.

- Compare your graph with your partner.

Discuss your graphs and make any changes, if necessary.

- Peter sketched the graph shown to represent the situation.

- Compare Peter's graph to your graph. Describe a feature Peter included on his graph that provides additional or different information than yours.

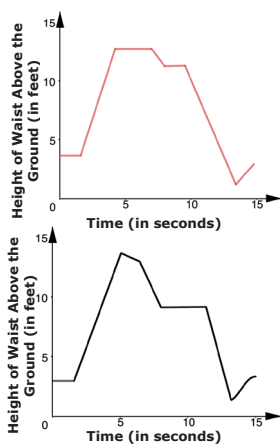
Answers will vary. Some possible differences include starting location, the increase in height at the end, or a different constant rate of change.

- What is the domain of Peter's graphical representation?

The domain of Peter's graphical representation is $0 \leq x \leq 15$.

- Discuss with your partner how the sketch of the graph would change if the y-axis measured "height of the man's feet above the ground (feet)" instead of "height of waist."

If the y-axis measured the "height of the man's feet above the ground (feet)" instead of "height of waist" the graph would change because it would start at 0 instead of starting above ground.



10.6.2 ACTIVITY: GRAPHING STORIES

TEACHER MOVES

Notice the intentional scaffolding for Story 1 to prime students for engaging in Stories 2-4 on their own in small groups or with partners.

- Play the video for Story 1 for the whole group.
- Ask students to imagine sliding down a playground slide and to show their distance above the ground as time passes using their fingers.
- Have students write their responses for questions 1 and 2 before a Turn and Talk with a partner for question 3.
- Students can then engage in all three parts for question 4 in pairs. Depending on student responses, the teacher can review responses and clarify misconceptions as a whole group or highlight strong connections students are making.
- Then release students in pairs or small groups to follow a similar process for Stories 2-4.
- Students can independently watch videos or view screenshots for all three stories before comparing with a partner, or partners can work together before comparing with someone else from around the room.

TEACHER MOVES

Anticipate possible differences in student responses and how they might inform different interpretations or highlight misconceptions.

Students may depict a y-intercept at zero, showing that he started "on the ground." (Incorrect axis label.)

Students may show the end of the slide and the initial height of the waist as equivalent. (This may be true depending on the height of the person, like a middle school student, for example.)

Students may not include time at which the height is not moving, like when the man is sitting, waiting to go down the slide.

QUESTIONING

Challenge students to make a connection between the "if" question here and the differences in the graphs of the dog run from the 10.6.1 Warm-Up. What would the dependent variable need to be for the graph to appear inverted from Peter's graph (like the 10.6.1 Warm-Up graphs appear inverted)?

10.6.2 ACTIVITY: GRAPHING STORIES (CONTINUED)

ENRICHMENT

- Monitor student responses for an increased rate when someone begins pouring the water while the faucet is running.
- What happens to the water level during the time someone is pouring water into the bowl?
- Is the water level increasing at the same rate or a faster rate?
- How do we represent a faster rate on the sketch of the graph?

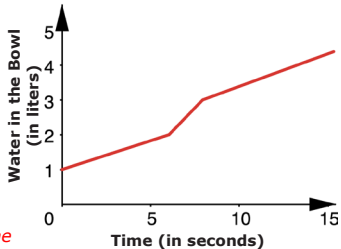
Use the screenshots and descriptions for three more stories to sketch a graph representing each story.



Story 2 – Filling with Water	
Screenshot	Description
	A bowl and measuring cup each contain some water. The bowl is filled with water at a constant rate from a faucet.
	After about 5 seconds, someone pours the water from the measuring cup into the bowl as the faucet continues to run.
	The faucet continues to run for the next several seconds.

5. Sketch the graph of the relationship between the amount of water in the bowl, in liters, and time, in seconds.

Student graphs will vary. This example is not the only solution to the problem.



TEACHER MOVES

The graph is discontinuous. Having students experience a discontinuous graph extends beyond the clarification statement. It is a type of function that students will experience on their mathematical journey and was included to build a student’s foundation and understanding of the real-world possibilities of such graphs.

ENRICHMENT

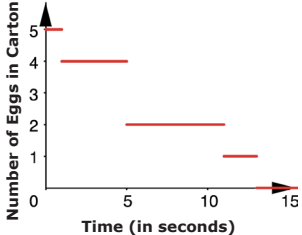
Students may have a difficult time representing the “jumps” when eggs are removed from the carton (piecewise function). Students might represent this as a continuous line with a slope. Reference points between whole numbers on their lines to guide their thinking.

In context, what does the point (2.5,5) mean? Is it possible to have 2.5 eggs in this context? So, how could we represent the function “jumping” from 4 eggs to 2 eggs?

Story 3 – Making Breakfast	
Screenshot	Description
	Several eggs are in an open egg carton.
	After 1 second, someone removes an egg.
	At 5 seconds, someone removes two more eggs.
	At 11 seconds, someone removes another egg.
	At 13 seconds, someone removes the last egg.

6. Sketch the graph of the relationship between the number of eggs in the carton and time, in seconds.

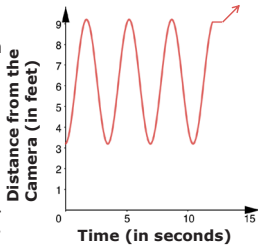
Student graphs will vary. This example is not the only solution to the problem.



10.6.2 ACTIVITY: GRAPHING STORIES
(CONTINUED)

Story 4 – Carousel Ride		
Screenshot	Description	
	A man is "riding" a carousel by pushing himself around with one foot. 	
	The screenshot shows the man after 1 second has passed.	
	The screenshot shows the man after 2 seconds have passed.	
	The man continues riding around and around on the carousel, then at 12 seconds, he gets off the carousel and walks away with his back to the camera.	

7. Sketch the graph of the relationship between the distance of the man from the camera, in feet, and time, in seconds.
Student graphs will vary. This example is not the only solution to the problem.



8. Compare your graphs for stories 2 and 4 with your partner. Discuss any differences and make changes to your graphs based on your discussion, if necessary.

TEACHER MOVES

Slides 10-12 in the Desmos Activity Builder provide real-time correct answers for each graph and story. Consider reviewing as a whole group and having students answer question 8 whole group as well for all three stories.

10.6.3 TRY IT: CREATE THE STORY

ENRICHMENT

Students are interpreting the winner as the line that “arrives” at 4 miles first as depicted in the graph.

- How would you know who finishes first?
- How long is the race? What distance (y-coordinate) represents the finish line?
- What does the x-coordinate at that point represent? How could you use that to determine who finished first?

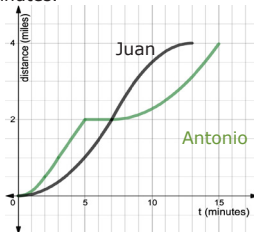
DIFFERENTIATE INSTRUCTION

Consider providing students an opportunity to use FlipGrid or other apps to actually record their race commentary to submit as a product of their learning.



10.6.3 Try It: Create the Story

1. Antonio and Juan are in a 4-mile bike race. The graph shows the distance of each racer, in miles, as a function of time, in minutes.



- a. Who won the race? Explain your reasoning.
Juan won the race because he finished it in less time.
- b. Imagine you were watching the race and had to announce it over the radio. Write a story describing the race as a sports commentator would.
Answers will vary. Possible answer: Antonio and Juan are beginning the race and Antonio with a quick pace takes the lead. We are 5 minutes into the race and it appears that Antonio is steadying his pace while Juan is keeping a slowly increasing speed. It looks like Juan is taking the lead as we reach the 7-minute mark. Antonio is speeding up, but it will not be enough with Juan winning the race at 13 minutes.



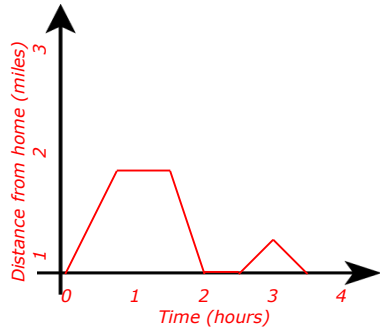
10.6.4 Wrap-Up: Cool Down

The story below describes Nima's Saturday afternoon.

"Nima was at home. He got on his bike and rode to his friend's house and stayed there for a while. Then, he rode home again. Later, he rode to the park. Then, he rode home again."

1. Sketch a graph of this story.

Student graphs will vary.



2. Label the axes of your graph with the names and units of measure for the two quantities being compared.
3. Based on your graph, is Nima's friend's house or the park closer to his house? Explain how you know.

Answers will vary. Based on the graph above the park is closer (answer will depend on whether the second part of the graph represents a distance closer to or farther from home). I know the park is closer because the graph shows Nima rode 2 miles to his friend's house and only 1 mile to the park.

10.6.4 WRAP-UP: COOL DOWN

TEACHER MOVES

Notice that correct responses will be completely determined by student sketches. Students may ask for specific values to make this determination.

Their precision in questions 1 and 2 will directly inform their unique correct answer in question 3.